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4 April 1986
SUPERSEDING
MIL-H-46354A(MU)
21 October 1974

MILITARY SPECIFICATION

HEADRESTS, OPTICAL INSTRUMENTS

This specification is approved for use within US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers headrests, browrests, and crash pads for use with Fire Control Instruments. These items collectively are referred to in this specification as headrests.

1.2 Classification. Headrests shall be of the following types as specified (see 6.2):

Type I - Encased mounting bracket
Type II - Adhered mounting bracket
Type III - Without mounting bracket

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament, Research and Development Center, ATTN: SMCAR-ESC-AS, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC: N/A
FSC 1240
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SPECIFICATIONS

Military

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection General Specifications For
MIL-P-14232	-	Parts, Equipment and Tools for Army Material, Packaging Of
MIL-I-45607	-	Inspection Equipment, Acquisition, Disposition and Maintenance Of

STANDARDS

Federal

FED-STD-595	-	Colors
FED-STD-601	-	Rubber Sampling and Testing

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-289	-	Visual Inspection Guide for Rubber Sheet Material
MIL-STD-293	-	Visual Inspection Guide for Cellular Rubber Items

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the document which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM

ASTM D471	-	Rubber Property, Effect of Liquid

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 29203).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

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2.3 Order or precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. Headrests furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time for opening of bids (see 4.2 and 6.3).

3.2 Construction. The headrests shall be Type I, Type II, or Type III constructed to the form and dimensions specified on the applicable drawing (see 6.2).

3.2.1 Type I. Type I headrests shall consist of a cellular elastomeric pad molded through and around a perforated mounting bracket and coated with a minimum 0.02 inch thick elastomeric skin.

3.2.2 Type II. Type II headrests shall consist of a molded cellular elastomeric pad adhered to a steel plate and the entire assembly coated with a minimum 0.02 inch thick elastomeric skin.

3.2.3 Type III. Type III headrests shall consist of cellular elastomeric pad coated with a minimum 0.02 inch thick elastomeric skin.

3.3 Materials. Natural rubber shall not be used for portions of the assembly exposed directly to the atmosphere.

3.3.1 Metal components. Metal components shall be in accordance with the applicable drawings and MIL-F-13926.

3.4.1 Coating to pad. The adhesive strength of the bond between the coating and the pad shall be greater than the cohesive strength of the pad or of the coating.

3.4.2 Pad to backplate, Type II only. The adhesive strength of the bond between the pad and the backplate shall be greater than the cohesive strength of the pad.

3.5 Coating

3.5.1 Tensile strength and elongation. The coating material shall have a minimum tensile strength of 300 psi and a minimum ultimate elongation of 200 per cent.

3.5.2 Fungus resistance. The coating shall be fungistatic, showing no evidence of fungus growth at the conclusion of 28 days exposure to fungal environment.

3.5.3 Oil resistance. The coating material shall comply with the oil resistance requirement, percent volume change of 0 to 120% when tested in accordance with 4.6.4.3.

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3.5.4 Color. The color of the coating material shall be black comparable to Color Chip No. 27038 or 37038 of FED-STD-595.

3.5.5 Workmanship. Workmanship shall be of a quality consistent with the highest commercial standards. All material shall be sound, of uniform quality and condition, and free from structural defects. The coating shall be free of runs, blisters, pock marks and surface voids or inclusions.

3.6 Flexibility. The flexibility of the headrests in terms of unit pressure required to produce a compression-deflection equal to 25 percent of the thickness of the pad, shall be 7 ± 3 pounds per square inch (psi) at room temperature, 21° ± 3° C (70° ± 8° F).

3.7 Resilience. The headrests shall return to within 95% of its original shape within one minute following flexibility testing at room temperature.

3.8 Accelerated aging. After exposure to a temperature of 70° ± 1° C (158° ± 3° F) for a period of 70 hours, the headrests shall not become tacky or deformed. Upon return to room temperature, flexibility of the pad shall be 7 ± 4 psi for 25 percent compression deflection.

3.9 Low temperature flexibility.

3.9.1 Type I. Type I specimens previously subjected to the accelerated aging test shall require a unit pressure of not more than 15 psi to attain 25 percent compression deflection at -40° ± 1° (-40° ± 1° F).

3.9.2 Type II & Type III. Type II and Type III specimens previously subjected to the accelerated aging test shall require a unit pressure of not more than 20 psi to attain 25 percent compression deflection at -40° + 1° C (-40° ± 1° F) after thermal stabilization at this temperature for 20 to 24 hours.

3.10 Toxicity and reaction with human skin. All portions of the fabricated headrest shall be inert and inherently incapable of causing a toxic or irritant effect as a result of contact with human skin.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TEST SPECIMEN
TYPE I CONSTRUCTION

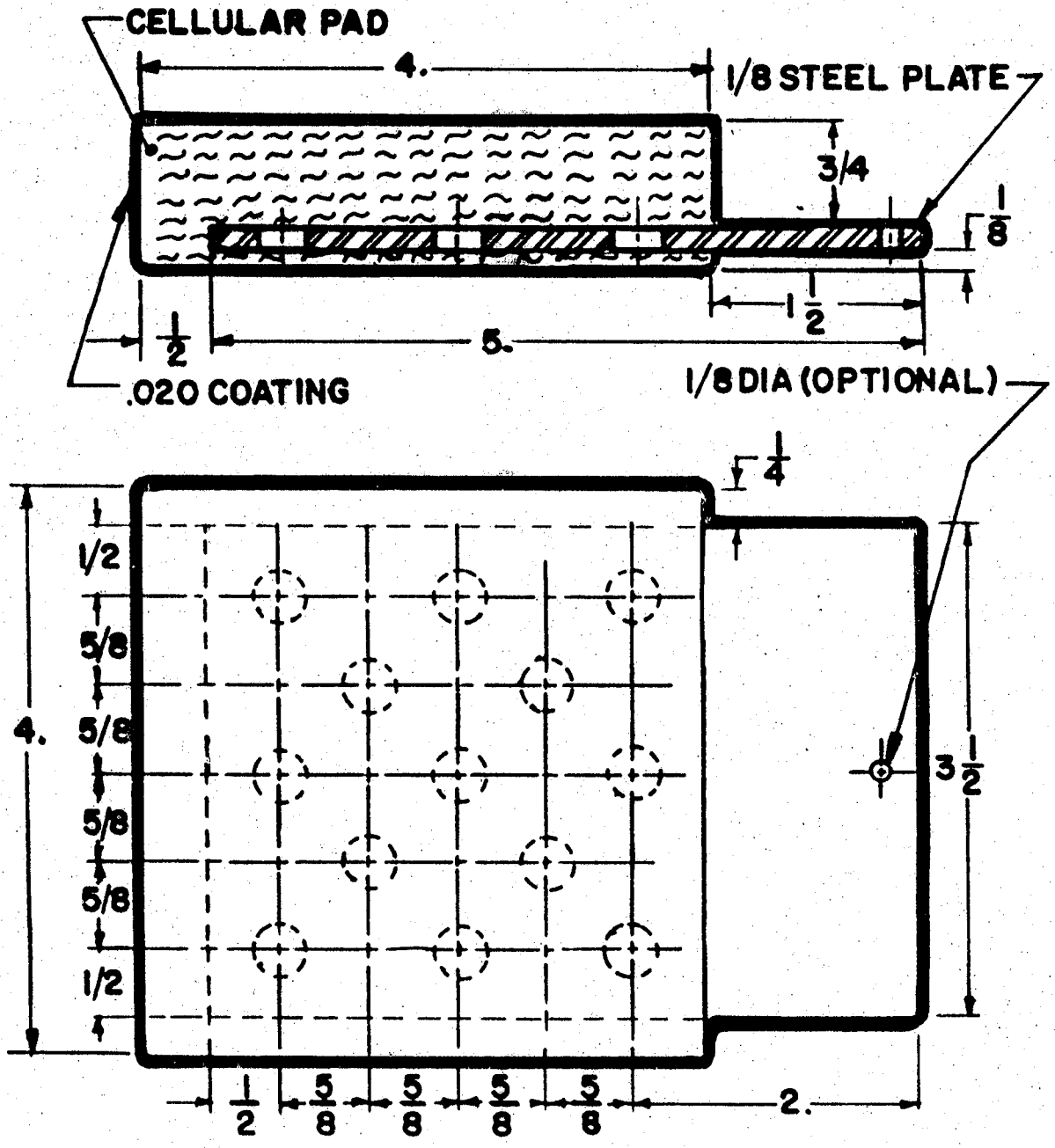


FIGURE 1

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TEST SPECIMEN
TYPE II CONSTRUCTION

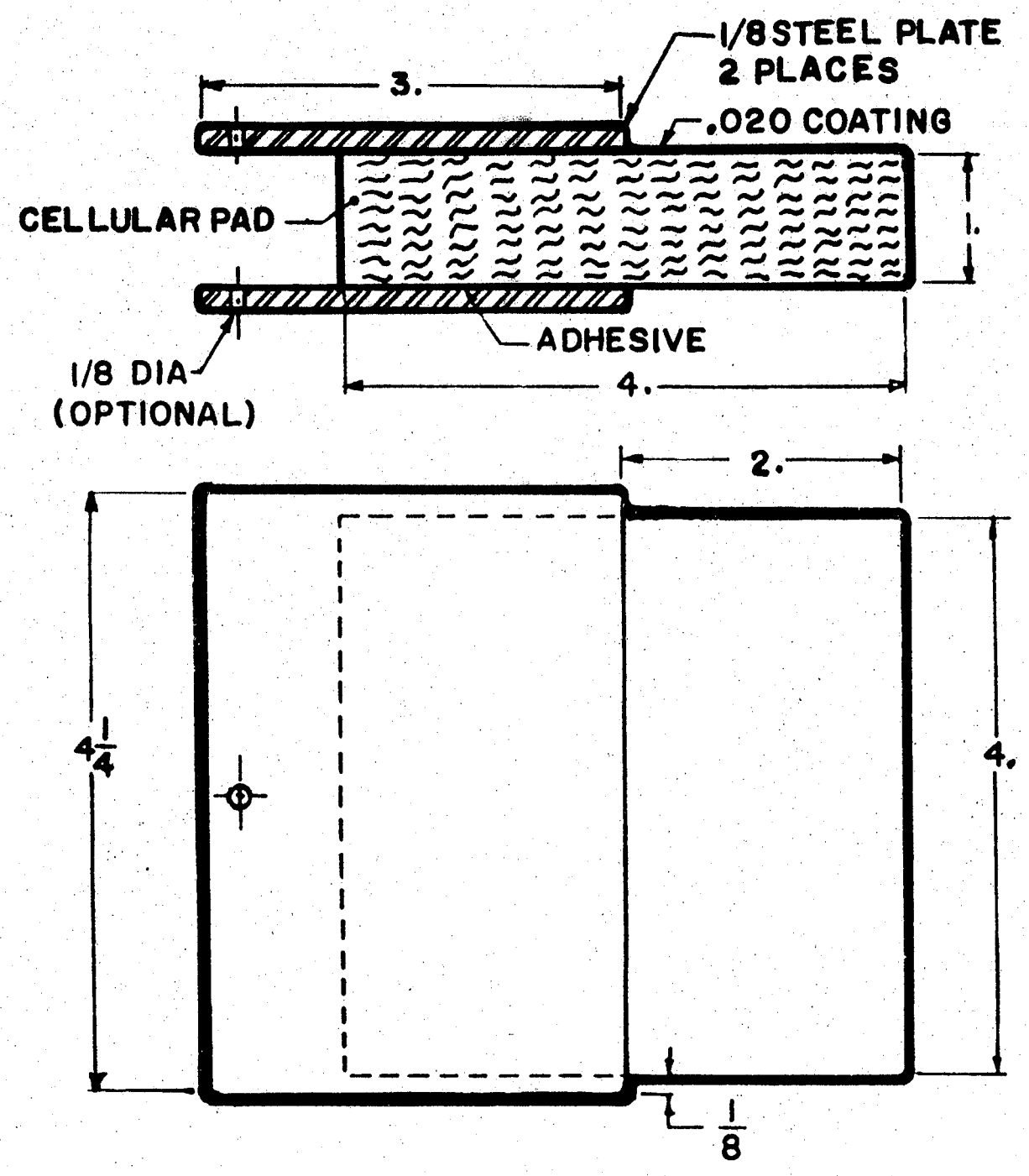


FIGURE II

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4.2 Classification of inspection.

a. Qualification inspection
b. Conformance inspection

4.3 Qualification inspection. A qualification sample of eight test specimens of each type specified shall be submitted for qualification testing in compliance with 3.1. Test specimens for Type I shall be in accordance with Figure I. Test specimens for Type II shall be in accordance with Figure II. Test specimens for Type III shall be in accordance with Figure II without the bonded steel plates. The samples shall be representative of the materials and construction of the headrest proposed to be furnished under contract. Qualification testing shall be conducted at a laboratory approved by the Government and shall consist of an examination for the characteristics specified in Table I in the order listed.

TABLE I. Qualification testing

CHARACTERISTIC	REQUIREMENT	TEST
1. Materials	3.3 & 3.3.1	4.6.2
2. Adhesive bond strength		
(a) Coating to pad	3.4.1	4.6.3.1
(b) Pad to backplate Type II only	3.4.2	4.6.3.2
3. Coating		
(a) Tensile strength and elongation	3.5.1	4.6.4.1
(b) Fungus resistance	3.5.2	4.6.4.2
(c) Oil resistance	3.5.3	4.6.4.3
(d) Color	3.5.4	4.6.4.4
(d) Workmanship	3.5.5	4.6.4.5
4. Flexibility	3.6	4.6.5
5. Resilience	3.7	4.6.6
6. Accelerated aging	3.8	4.6.7
7. Low temperature flexibility		
(a) Type I	3.9.1	4.6.8
(b) Type II & III	3.9.2	4.6.8
8. Toxicity & reaction to human skin	3.10	4.6.9

NOTE: Two specimens shall be selected for the fungus resistance test (see 4.6.4.2) from each of the eight specimens submitted. This shall be the only test required for these two specimens.

4.4 Conformance inspection.

4.4.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.4.2 Classification of examination and tests. Examinations, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table II herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table II shall constitute the minimum inspection to be performed by the supplier after qualification approval and prior to Government acceptance or rejection by item or lot.

TABLE II. Conformance testing & classification of defects

Use Inspection Level II of Table I with sampling Plan Table II-A of MIL-STD-105.

CRITICAL: None Defined

MAJOR: AQL 1.0

CHARACTERISTIC	REQUIREMENT	TEST
101. Construction		
(a) Type I	3.2.1	4.6.1
(b) Type II	3.2.2	4.6.1
(c) Type III	3.2.3	4.6.1
102. Flexibility	3.6	4.6.5
103. Resilience	3.7	4.6.6
104. Workmanship	3.5.5	4.6.4.5
MINOR: AQL 4.0		
201. Color	3.5.4	4.6.4.4
202. Identification Marking		5.1

NOTE: The above tests shall be conducted at 21° ± 2° C (70° ± 5°F).

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmission of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4.4 Certification. Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

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CHARACTERISTICS	REQUIREMENT	TEST
Materials	3.3	4.6.2
Fungus resistance	3.5.2	4.6.4.2
Accelerated aging	3.8	4.6.7
Low temperature flexibility	3.9	4.6.8
Toxicity	3.10	4.6.9

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Construction. The headrest shall be subjected to a visual, tactile and dimensional inspection for conformance with the construction and configuration requirements of 3.2. Standard measuring equipment shall be used for dimensional inspections.

4.6.2 Materials. The headrest shall be visually examined for compliance with 3.3. The government shall reserve the right to require a chemical analysis of materials or certification thereof, to insure quality of material in accordance with respective drawing of specification requirements specified herein.

4.6.3 Adhesive bond strength.

4.6.3.1 Coating to pad. The test specimen shall be a sample specimen

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of either Type I, Type II, or Type III construction, as required, and shall be a specimen not previously used in other tests. The test shall be performed using a suitable tensile test device. Two parallel cuts one inch apart shall be made through the coating and one end of the strip thus formed shall be cut free and pulled up perpendicular to the specimen surface sufficient to form a gripping tab for the pull test. When pulled to failure at a rate of five inches per minute at 25° ± 1° C (77° ± 2° F), the rupture shall occur in either the pad or the coating and not in the bond (adhesive).

4.6.3.2 Pad to backplate. The test specimen shall be a sample specimen of Type II only and shall be a specimen not previously used in other tests. This test shall be performed using a suitable tensile test device. The headrest shall be pulled perpendicular to each backplate surface. Pulled to failure at a rate of five inches per minute, while at 25° ± 1°C (77° ± 2° F) a rupture shall occur in the skin or padding and not in the skin-to-backplate bond.

4.6.4 Coating.

4.6.4.1 Tensile strength and elongation. This test shall be performed in accordance with Method 4111 of FED-STD-601. Sample test pieces shall be obtained by stripping the coating from the backplates or pads on the test specimens used in the adhesive bond strength test of 4.6.3. The tensile strength and elongation of the coating shall meet the requirements of 3.5.1.

4.6.4.2 Fungus resistance. Two headrests (selected from the eight qualification samples submitted) shall be tested for compliance with 3.9.2 using Method 508.3 of MIL-STD-810.

4.6.4.3 Oil resistance. The coating material shall comply with the requirement stated in 3.5.3 when tested at 100° ± 2° C (212° ± 5° F) using #3 oil for 70 hours min. in accordance with ASTM D471.

4.6.4.4 Color. The test specimen or headrest shall be visually examined for correctness of color. Unless otherwise specified the color shall be a black comparable to Color Chips No. 27038 or No. 37038 of FED-STD-595.

4.6.4.5 Workmanship. The test specimens or headrests shall be visually inspected for conformance to the requirements of 3.5.5. The specimens or headrests shall be free from the specified defects. MIL-STD-289 and MIL-STD-293 shall be used as an inspection guide in this examination.

4.6.5 Flexibility. The load applied to the specimen or headrest to determine compliance with 3.4 shall be made with a compressiosn testing device. The device shall be equipped with a circular pressure foot with a surface contact area of 1.0 square inch. For contact areas that do not have this size of flat area, a pressure foot with an area of 0.5 square inch minimum shall be used. The device shall be capable of exerting a force of ± 3 pounds per square inch (psi) on the specimen through a force gage with a minimum range of 10 psi, with an accuracy of 1/2 psi of full scale. The largest flat area of uniform thickness on the specimen shall be used as the test location. Apply the pressure gradually and in a perpendicular plane

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until the pressure foot has traveled 25 ± 2 percent of the free height of the specimen or headrest. The force required for this compression shall be read from the force gage and shall not exceed the value specified. Force readings shall be taken 15 to 30 seconds after initial application of the force.

4.6.6 Resilience. This test shall be conducted concurrent with the flexibility test. Upon removal of the applied load the specimen or headrest shall return to within 95 percent of its original shape within the time limit specified. Any deformation shall be measured with standard measuring equipment.

4.6.7 Accelerated aging. The aging test shall be performed in heating chamber equipped with automatic temperature controls capable of maintaining the chamber at the operating temperature specified and shall be suitably baffled to prevent dead spots and local overheating. The temperature variation throughout the chamber shall not exceed 1° C (3° F). The heating medium shall be air, thoroughly circulated by mechanical means at atmospheric pressure. The source of heat shall be located in the air supply outside the chamber proper. Provisions shall be made for the vertical suspension of specimens without contact with each other or the sides of the chamber. A recording thermometer shall be located in the upper central portion of the chamber near the center of the specimens or headrests. The specimen or headrest shall be placed in the chamber, after the chamber has been pre-heated to a temperature of 70° ± 1° C (158° ± 3° F), and shall be kept in the chamber at that temperature for a period of 70 hours. At the end of the 70 hour period, the specimen or headrest shall be removed from the chamber and subjected to a tactile and visual examination at room temperature 21° ± 3° C(70° ± 8° F) on a flat surface for a period of at least two hours. The specimen or headrest shall again be subjected to the flexibility test specified in 4.6.3. The force required for this compression deflection shall not exceed the value specified.

4.6.8 Low temperature flexibility (Type I, Type II, or Type III). This test shall be performed in a cold chamber maintained at the specified low temperature. Subsequent to the oven aging test the specimen or headrest shall be placed in the cold chamber for a period of 20 to 24 hours. Immediately following this time cycle, and while the specimens are in the chamber, they shall be subjected to the flexibility test. The force required for a 25 percent compression deflection shall not exceed the value specified for the type of construction under test. The device and method of application used to perform the flexibility test of 4.6.3 shall be used for this test.

4.6.9 Toxicity and reaction to human skin. This test shall be a physical and visual test to determine compliance with 3.3.10. The headrest or specimen shall not cause a burning or itching sensation when held against the human skin at the forehead for a period of not less than three minutes in the ambient temperature, 34° ± 1° C (93° ± -3° F). Also the material shall not become tacky after contact with the human skin for the duration specified above. In addition to the above, the contractor shall certify that the material used is not considered toxic, or known to cause dermatitis or other inflammatory reaction when in contact with human skin.

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5. PACKAGING

5.1 Packaging and packing. The headrests shall be cleaned, preserved, packaged, packed and marked in accordance with MIL-P-14232 and the applicable packaging data sheet. The level of protection and data sheet shall be as specified in the procurement document (see 6.2).

6. NOTES

6.1 Intended use. Headrests manufactured to this specification are intended for Fire Control application in periscopes, telescopes, range finders, etc.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type of headrest and applicable drawings.
c. Applicable stock number.
d. Applicable packaging data sheet and level of protection.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is US Army Armament, Research, Development and Engineering Center, ATTN: SMCAR-ESC-AS, Dover, NJ 07801-5001.

Custodian:
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